[26-PATCHALIAS] Structural Aliasing in Patch-Based Time-Series Foundation Models

Project Co-Supervision

This coursework is co-supervised with Prof. Benini and it is joint with the Generative AI for Media course. It is limited to a single group of maximum size 2. The final deliverable must be of maximum 7 000 words (plus 10% grace).

Patch-based tokenisation is widely adopted in time-series foundation models to reduce sequence length and improve computational efficiency. However, this design choice may introduce architectural distortions that are not captured by classical sampling theory. In particular, when the temporal support of a patch is commensurate with the period of an input signal, successive patches may become identical or exhibit strong redundancy. This can make certain frequencies poorly observable to the model despite being fully representable under the Nyquist–Shannon sampling theorem. The resulting phenomenon can be interpreted as a form of *structural aliasing*, induced by the interaction between signal periodicity and tokenisation rather than by undersampling.

The problem is not only signal-theoretic, but also semantic. In many digital transformation settings, time-series signals are interpreted through domain concepts such as events, states, anomalies, operating regimes, or control conditions. Students should therefore consider whether structural aliasing distorts not only the observability of frequencies, but also the semantic interpretation of the signal. Where appropriate, this semantic layer should be formalised through ontologies or comparable knowledge representations, so that the link between signal structure, learned representation, and domain meaning becomes explicit. This is especially relevant in agentic AI settings, where a model may be embedded in an agent that monitors environments, explains detected conditions, supports decisions, or triggers actions on the basis of time-series inputs.

This coursework aims to develop a formal and falsifiable understanding of structural aliasing in patch-based time-series models. The work must combine theoretical derivation, empirical validation, and principled statistical analysis. In particular, the group must:

1. Formally model patch-based tokenisation as an operator acting on discrete-time signals. Derive the conditions under which two consecutive patches become identical or indistinguishable, and characterise the resulting loss of observability in the token sequence.
2. Establish the relationship between sampling frequency (f_s), patch size (P), stride (S), and signal frequency (f). In particular, derive the class of frequencies for which phase-locking with the patch grid occurs, and analyse whether these induce degeneracies or

invariances in the token space.

3. Define the semantic layer of the chosen application setting by identifying the relevant concepts, states, and relations through an ontology or a comparable formal representation. The coursework must discuss how structural aliasing at the signal and token levels may affect the correct recognition of these domain-level concepts.
4. Clearly define a set of falsifiable hypotheses regarding structural aliasing. These must include, at minimum: (i) the existence of frequency-dependent blind spots at harmonics of a patch-induced frequency; (ii) the dependence of the effect on temporal alignment (phase shifts); (iii) the impact of patch size, stride, and overlap on the phenomenon.
5. Design controlled experiments using synthetic signals (*e.g.*, sinusoids and their extensions) to test the hypotheses. The experiments must isolate architectural effects from classical sampling effects and must include systematic variation of patch parameters and temporal offsets.
6. Evaluate the impact of the phenomenon both at the behavioural level (*e.g.*, prediction accuracy, spectral reconstruction) and, where possible, at the representational level (*e.g.*, probing or equivalent analysis of internal states).
7. Analyse the results using Bayesian statistical methods. In particular, students must specify prior assumptions where appropriate, quantify uncertainty in the estimated effects, and compare competing hypotheses or mitigation claims using Bayesian inference rather than relying only on point estimates or null-hypothesis significance testing.
8. Investigate at least one mitigation strategy and provide a principled explanation of why it does or does not address the identified structural aliasing effect.
9. Discuss how the developed techniques could be used in an agentic AI system operating over time-series data. In particular, the final report must comment on how an agent could represent, monitor, explain, and act upon structural aliasing risks, and on how semantic and Bayesian analyses could support more reliable decision-making and human-facing explanations.

Joint Computer Security / Digital Transformation

For Computer Security applicants, the coursework must include a formal risk assessment grounded in the NIST AI Risk Management Framework (AI RMF), treating structural aliasing as a potential source of systematic vulnerability in time-series models. The final report must discuss plausible threat models, attack surfaces, and realistic consequences in security-relevant contexts such as anomaly detection, monitoring, and cyber-physical systems, and must evaluate whether the proposed mitigation strategies reduce these risks or leave significant residual exposure. In this case, the final output is limited to 8 000 words (plus 10% grace).

Readings

- Ansari, Abdul Fatir, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Haixu Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sairam Praneeth Arango, Shubham Kapoor, Jan Zschiegner, David C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Matthias Bohlke-Schneider, and Yuyang Wang. "Chronos: Learning the Language of Time Series." arXiv preprint arXiv:2403.07815, 2024.
- Voita, Elena, and Ivan Titov. "Information-Theoretic Probing with Minimum Description Length." In *Proceedings of EMNLP*, 2020.
- Belrose, Nora, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. "LEACE: Perfect Linear Concept Erasure in Closed Form." In *Advances in Neural Information Processing Systems*, 2023.
- Oppenheim, Alan V., and Ronald W. Schafer. *Discrete-Time Signal Processing*. 3rd ed. Upper Saddle River, NJ: Pearson, 2010.
- National Institute of Standards and Technology. "Artificial Intelligence Risk Management Framework (AI RMF 1.0)." 2023.